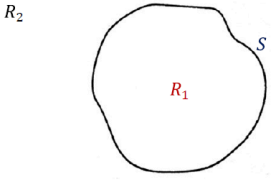
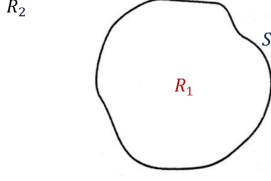


Electrostatics VS Magnetostatics

Electrostatics	Magnetostatics
In free space	
 Consider a volume of space R_1, with boundary S that separates it from a volume of space R_2. Suppose there is an electrostatic field in R_1, and there is another electrostatic field in R_2.	 Consider a volume of space R_1, with boundary S that separates it from a volume of space R_2. Suppose there is a magnetostatic field in R_1, and there is another magnetostatic field in R_2.
The electric scalar potential V_1 in R_1 is determined by the Poisson equation, $\nabla^2 V_1 = -\frac{1}{\epsilon_0} \rho_1$ where ρ_1 is the electric charge per unit volume in R_1. While, the electric scalar potential V_2 in R_2 is determined by the Poisson equation, $\nabla^2 V_2 = -\frac{1}{\epsilon_0} \rho_2$ where ρ_2 is the electric charge per unit volume in R_2. ∇^2 in per meter per meter, $1/m^2 = m^{-2}$ V in energy per unit charge, $V = J/C = \text{kgm}^2\text{s}^{-3}\text{A}^{-1}$ ϵ in capacitance per unit length, $F/m = \text{A}^2\text{s}^4\text{kg}^{-1}\text{m}^{-3}$ Capacitance in $F = C/V = \text{A}^2\text{s}^4\text{kg}^{-1}\text{m}^{-2}$ $\epsilon_0 \sim 8.854 \times 10^{-12} \text{ A}^2\text{s}^4\text{kg}^{-1}\text{m}^{-3}$ ρ in charge per unit volume, $C/m^3 = \text{Asm}^{-3}$	The magnetic vector potential A_1 in R_1 is determined by the Poisson equation, $\nabla^2 A_1 = -\mu_0 J_1,$ where J_1 is the electric current per unit area in R_1, and A_1 satisfies $\nabla \cdot A_1 = 0$. While, the magnetic vector potential A_2 in R_2 is determined by the Poisson equation, $\nabla^2 A_2 = -\mu_0 J_2,$ where J_2 is the electric current per unit area in R_2, and A_2 satisfies $\nabla \cdot A_2 = 0$. $\nabla \cdot A = 0$ is chosen to be like so due to Gauge Fixing ∇^2 in per meter per meter, $1/m^2 = m^{-2}$ A in force per unit current $\text{NA}^{-1} = \text{Tm} = \text{Wbm}^{-1} = \text{kgms}^{-2}\text{A}^{-1}$ μ in inductance per unit length, $\text{H/m} = \text{N/A}^2 = \text{kgms}^{-2}\text{A}^{-2}$ Inductance in $\text{H} = \text{kgm}^2\text{s}^{-2}\text{A}^{-2}$ $\mu_0 = 4\pi \times 10^{-7} \text{ kgms}^{-2}\text{A}^{-2}$ J in current per unit area, $\text{A/m}^2 = \text{Am}^{-2}$
The electrostatic field in R_1 can be described by the electric scalar potential V_1, or electric field intensity $E_1 = -\nabla V_1,$ or electric flux density $D_1 = \epsilon_0 E_1.$ This has energy and the energy per unit volume is given by $u_E = \frac{1}{2} \epsilon_0 E_1 \cdot E_1 = \frac{1}{2} D_1 \cdot E_1$ ($F_{\text{electric}} = qE$) The electrostatic field in R_2 can be described by the electric scalar potential V_2, or electric field intensity $E_2 = -\nabla V_2,$ or electric flux density $D_2 = \epsilon_0 E_2.$ This has energy and the energy per unit volume is given by $u_E = \frac{1}{2} \epsilon_0 E_2 \cdot E_2 = \frac{1}{2} D_2 \cdot E_2$ $E = -\text{ve gradient of a scalar potential} = -\nabla V$ $\Rightarrow E$ is a gradient field $\Rightarrow \text{curl of } E = \nabla \times E = 0 \text{ vector}$ E in force per unit charge, $\text{N/C} = \text{kgms}^{-3}\text{A}^{-1}$ ∇ in per meter, $1/m = m^{-1}$ V in energy per unit charge, $V = J/C = \text{kgm}^2\text{s}^{-3}\text{A}^{-1}$ D in $\text{C/m}^2 = \text{Asm}^{-2}$ ϵ in capacitance per unit length, $\text{F/m} = \text{A}^2\text{s}^4\text{kg}^{-1}\text{m}^{-3}$ E in force per unit charge, $\text{N/C} = \text{kgms}^{-3}\text{A}^{-1}$ u_E in energy per unit volume, $\text{J/m}^3 = \text{kgm}^{-1}\text{s}^{-2}$	The magnetostatic field in R_1 can be described by the magnetic vector potential A_1, or magnetic flux density $B_1 = \nabla \times A_1,$ or magnetic field intensity $H_1 = \frac{1}{\mu_0} B_1$ This has energy and the energy per unit volume is given by $u_B = \frac{1}{2\mu_0} B_1 \cdot B_1 = \frac{1}{2} H_1 \cdot B_1$ ($F_{\text{magnetic}} = qv \times B$) The magnetostatic field in R_2 can be described by the magnetic vector potential A_2, or magnetic flux density $B_2 = \nabla \times A_2,$ or magnetic field intensity $H_2 = \frac{1}{\mu_0} B_2$ This has energy and the energy per unit volume is given by $u_B = \frac{1}{2\mu_0} B_2 \cdot B_2 = \frac{1}{2} H_2 \cdot B_2$ $B = \text{curl of } A = \nabla \times A$ B in $\text{T} = \text{Wb/m}^2 = \text{kgs}^{-2}\text{A}^{-1}$ ∇ in per meter, $1/m = m^{-1}$ A in force per unit current $\text{NA}^{-1} = \text{Tm} = \text{Wbm}^{-1} = \text{kgms}^{-2}\text{A}^{-1}$ H in $\text{A/m} = \text{Am}^{-1}$ μ in inductance per unit length, $\text{H/m} = \text{N/A}^2 = \text{kgms}^{-2}\text{A}^{-2}$ B in $\text{T} = \text{Wb/m}^2 = \text{kgs}^{-2}\text{A}^{-1}$ u_B in energy per unit volume, $\text{J/m}^3 = \text{kgm}^{-1}\text{s}^{-2}$

In addition, everywhere on the boundary S , we have

$$V_1 = V_2,$$

and

$$\mathbf{E}_{1t} = \mathbf{E}_{2t},$$

$$\mathbf{a}_{1n} \cdot \epsilon_0 \mathbf{E}_1 + \mathbf{a}_{2n} \cdot \epsilon_0 \mathbf{E}_2 = \rho_s,$$

where ρ_s is the electric charge per unit area on the boundary S .

- Tangential components of $\mathbf{E}$ in R_1 and R_2 are equal
- The amount of normal component of $\mathbf{E}$ in R_1 and R_2 determines the surface charge density
 - if no surface charge density, normal are of $\mathbf{E}$ in R_1 and R_2 are in the same direction and has the same magnitude

unit vector is unitless, purely for the direction
 ϵ in capacitance per unit length, $\text{F/m} = \text{A}^2 \text{s}^4 \text{kg}^{-1} \text{m}^{-3}$
 $\mathbf{E}$ in force per unit charge, $\text{N/C} = \text{kgms}^{-3} \text{A}^{-1}$
 ρ_s is charge per unit area, $\text{C/m}^2 = \text{Asm}^{-2}$

In addition, everywhere on the boundary S , we have

$$\mathbf{a}_{1n} \cdot \mathbf{B}_1 + \mathbf{a}_{2n} \cdot \mathbf{B}_2 = 0,$$

and

$$\mathbf{a}_{1n} \times \frac{1}{\mu_0} \mathbf{B}_1 + \mathbf{a}_{2n} \times \frac{1}{\mu_0} \mathbf{B}_2 = \mathbf{J}_s$$

where $\mathbf{J}_s$ is the electric current per unit length on the boundary S .

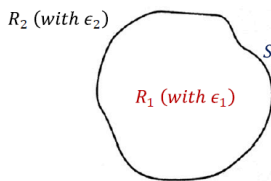
- Normal components of $\mathbf{B}$ in R_1 and R_2 are equal (not opposite as normal vector are opposite)
- $\mathbf{J}_s$ is perpendicular to both normal and $\mathbf{B}$, the amount of $\mathbf{B}$ perpendicular to normal and $\mathbf{B}$ determines how much current is flowing in the direction of $\mathbf{J}_s$
- $\mathbf{J}_s$ = current per unit length, length perpendicular to current flow

unit vector is unitless, purely for the direction
 μ in inductance per unit length, $\text{H/m} = \text{N/A}^2 = \text{kgms}^{-2} \text{A}^{-2}$
 $\mathbf{B}$ in $\text{T} = \text{Wb/m}^2 = \text{kgs}^{-2} \text{A}^{-1}$
 $\mathbf{J}_s$ is current per unit length, A/m

In Dielectric Material

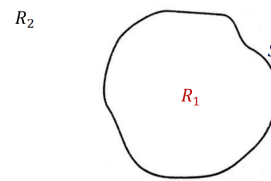
- if external $\mathbf{E}$ is present, the surface can have bound surface charge, but there is no free surface charge, unless is something introduced in by other means
- free we take the material $\epsilon \mu$
- total = free + bound (polarised) we take free space $\epsilon \mu$

In Magnetic Material



Consider a volume of dielectric material R_1 (e.g., glass), with boundary S that separates it from a volume of dielectric material R_2 (e.g., water).

Suppose there is an electrostatic field in R_1 , and there is another electrostatic field in R_2 .



Consider a volume of magnetic material R_1 (e.g., aluminium), with boundary S that separates it from a volume of magnetic material R_2 (e.g., water).

Suppose there is a magnetostatic field in R_1 , and there is another magnetostatic field in R_2 .

The electric scalar potential V_1 in R_1 is determined by the Poisson equation,

$$\nabla^2 V_1 = -\frac{1}{\epsilon_1} \rho_{\text{free},1} \quad \left(\text{compare } \nabla^2 V_1 = -\frac{1}{\epsilon_0} \rho_1 \right)$$

where $\rho_{\text{free},1}$ is the free electric charge per unit volume in R_1 . While, the electric scalar potential V_2 in R_2 is determined by the Poisson equation,

$$\nabla^2 V_2 = -\frac{1}{\epsilon_2} \rho_{\text{free},2} \quad \left(\text{compare } \nabla^2 V_2 = -\frac{1}{\epsilon_0} \rho_2 \right)$$

where $\rho_{\text{free},2}$ is the free electric charge per unit volume in R_2 .

7

The magnetic vector potential $\mathbf{A}_1$ in R_1 is determined by the Poisson equation,

$$\nabla^2 \mathbf{A}_1 = -\mu_1 \mathbf{J}_{\text{free},1} \quad \left(\text{compare } \nabla^2 \mathbf{A}_1 = -\mu_0 \mathbf{J}_1 \right)$$

where $\mathbf{J}_{\text{free},1}$ is the free electric current per unit area in R_1 , and $\mathbf{A}_1$ satisfies $\nabla \cdot \mathbf{A}_1 = 0$. While, the magnetic vector potential $\mathbf{A}_2$ in R_2 is determined by the Poisson equation,

$$\nabla^2 \mathbf{A}_2 = -\mu_2 \mathbf{J}_{\text{free},2} \quad \left(\text{compare } \nabla^2 \mathbf{A}_2 = -\mu_0 \mathbf{J}_2 \right)$$

where $\mathbf{J}_{\text{free},2}$ is the free electric current per unit area in R_2 , and $\mathbf{A}_2$ satisfies $\nabla \cdot \mathbf{A}_2 = 0$.

$\nabla \cdot \mathbf{A} = 0$ is chosen to be like so due to Gauge Fixing

The electrostatic field in R_1 can be described by the electric scalar potential V_1 , or electric field intensity

$$\mathbf{E}_1 = -\nabla V_1,$$

or electric flux density

$$\mathbf{D}_1 = \epsilon_1 \mathbf{E}_1.$$

This has energy and the energy per unit volume is given by

$$u_E = \frac{1}{2} \epsilon_1 \mathbf{E}_1 \cdot \mathbf{E}_1 = \frac{1}{2} \mathbf{D}_1 \cdot \mathbf{E}_1$$

The electrostatic field in R_2 can be described by the electric scalar potential V_2 , or electric field intensity

$$\mathbf{E}_2 = -\nabla V_2,$$

or electric flux density

$$\mathbf{D}_2 = \epsilon_2 \mathbf{E}_2.$$

This has energy and the energy per unit volume is given by

$$u_E = \frac{1}{2} \epsilon_2 \mathbf{E}_2 \cdot \mathbf{E}_2 = \frac{1}{2} \mathbf{D}_2 \cdot \mathbf{E}_2$$

$\mathbf{E} = -\text{ve gradient of a scalar potential} = -\nabla V$
 $\Rightarrow \mathbf{E}$ is a gradient field

The magnetostatic field in R_1 can be described by the magnetic vector potential $\mathbf{A}_1$, or magnetic flux density

$$\mathbf{B}_1 = \nabla \times \mathbf{A}_1,$$

or magnetic field intensity

$$\mathbf{H}_1 = \frac{1}{\mu_1} \mathbf{B}_1$$

This has energy and the energy per unit volume is given by

$$u_B = \frac{1}{2\mu_1} \mathbf{B}_1 \cdot \mathbf{B}_1 = \frac{1}{2} \mathbf{H}_1 \cdot \mathbf{B}_1$$

5

The magnetostatic field in R_2 can be described by the magnetic vector potential $\mathbf{A}_2$, or magnetic flux density

$$\mathbf{B}_2 = \nabla \times \mathbf{A}_2,$$

or magnetic field intensity

$$\mathbf{H}_2 = \frac{1}{\mu_2} \mathbf{B}_2$$

This has energy and the energy per unit volume is given by

$$u_B = \frac{1}{2\mu_2} \mathbf{B}_2 \cdot \mathbf{B}_2 = \frac{1}{2} \mathbf{H}_2 \cdot \mathbf{B}_2$$

6

$\Rightarrow \text{curl of } \mathbf{E} = \nabla \times \mathbf{E} = 0 \text{ vector}$ $\epsilon_r = \frac{\epsilon}{\epsilon_0}$ ϵ_r must be more than 1 - Higher means polarise more, easier to polarise	$\mathbf{B} = \text{curl of } \mathbf{A} = \nabla \times \mathbf{A}$ $\mu_r = \frac{\mu}{\mu_0}$ $\mu > \mu_0$ for paramagnetism - Bend away from normal, with 90° being the asymptote $\mu = \mu_0$ for air/free space - No bending $\mu < \mu_0$ for diamagnetism - Bend towards the normal, with 0° being the asymptote - With strongly magnetic materials, the μ is big, $\mathbf{B}$ bends away from normal in iron - iron, which is ferromagnetic, it's taken as a simple magnetic material of PC2020 - Higher means magnetise more, easier to magnetise
<i>In addition, everywhere on the boundary S, we have</i> $V_1 = V_2,$ <i>and</i> $\mathbf{E}_{1t} = \mathbf{E}_{2t},$ $\mathbf{a}_{1n} \cdot \epsilon_1 \mathbf{E}_1 + \mathbf{a}_{2n} \cdot \epsilon_2 \mathbf{E}_2 = \rho_{\text{free},s},$ <i>where $\rho_{\text{free},s}$ is the free electric charge per unit area on the boundary S.</i> Bending at the Boundary: Given R_1 with E_1, ϵ_1 and R_2 with ϵ_2 find E_2 - the tangential component must be the same - because for static electric field, the curl at all points must be 0 - given free surface charge = 0, the normal component of $E_2 = \epsilon_1/\epsilon_2 \cdot$ normal component of E_1 - if let's say $\epsilon_2 > \epsilon_1$, must bent away from normal, means with the same the tangential component, the normal component must be smaller - this is because higher permittivity, polarise more, weaker field	<i>In addition, everywhere on the boundary S, we have</i> $\mathbf{a}_{1n} \cdot \mathbf{B}_1 + \mathbf{a}_{2n} \cdot \mathbf{B}_2 = 0,$ <i>and</i> $\mathbf{a}_{1n} \times \frac{1}{\mu_1} \mathbf{B}_1 + \mathbf{a}_{2n} \times \frac{1}{\mu_2} \mathbf{B}_2 = \mathbf{J}_{\text{free},s}$ <i>where $\mathbf{J}_{\text{free},s}$ is the free electric current per unit length on the boundary S.</i> Bending at the Boundary: Given R_1 with B_1, μ_1 and R_2 with μ_2 find B_2 - the normal component must be the same - given free surface current = 0, the tangential component of $B_2 = \mu_2/\mu_1 \cdot$ tangential component of B_1 - if let's say $\mu_2 > \mu_1$, must bent away from normal, means with the same the normal component, the tangential component must be bigger - this is because higher permeability, magnetise more, stronger field
In summary: electric permittivity, charge per unit volume and charge per unit area will change to only account for the contribution of effect due to the free charge.	In summary: magnetic permittivity, current per unit area and current per unit length will change to only account for the contribution of effect due to the free current.
Effect in an electrostatic field is that particle will get polarised and align themselves in the field - Air with Wimshurst machine - But electric force $\gg$ magnetic force, breakdown of air is possible, electrons are ripped away from nucleus	Effect in an magnetostatic field is that particle will get magnetised and align themselves in the field - Iron filling with permanent magnet - But magnetic force $\ll$ electric force, breakdown of iron filling not possible
Capacitor	Inductor
Parallel Plate Capacitor, surface area S, distance apart, d	Parallel Plate Inductor, length l (current direction), width W, distance apart d

$$C_{free\ space} = \frac{\epsilon_0 S}{d}$$

$$C_{\epsilon_r} = \frac{\epsilon S}{d} = \frac{\epsilon_r \epsilon_0 S}{d}$$

$$C_{\epsilon_r} = \epsilon_r C_{free\ space}$$

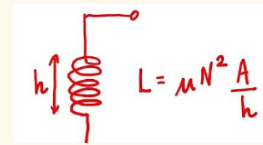
$$C = \frac{|Q_{free}|}{\Delta V}$$

- The region with larger ϵ , polarises more, hence will have lower E
 - Permittivity and Polarisation works against the E applied
 - larger ϵ , bigger capacitance
- Total energy = $1/2 C V^2$

$$L_{free\ space} = \frac{\mu_0 l d}{W}$$

$$L_{\mu_r} = \frac{\mu l d}{W} = \frac{\mu_r \mu_0 l d}{W}$$

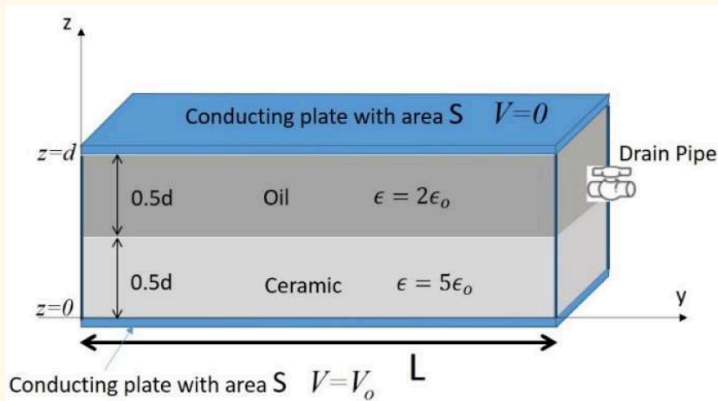
$$L_{\mu_r} = \mu_r L_{free\ space}$$



$$L = \frac{|\Phi|}{I}$$

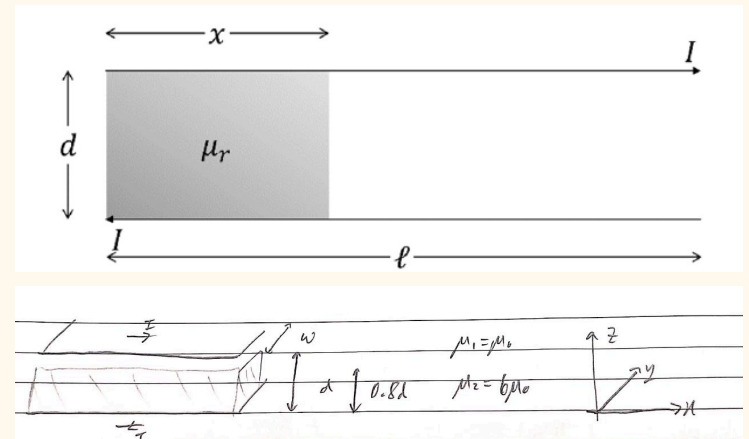
- The region with larger μ , magnetises more, hence will have higher B
 - Permeability and Magnetisation works together with the B applied
- Using total energy with $1/2 L I^2$ is easier

Parallel Plate Capacitor in series



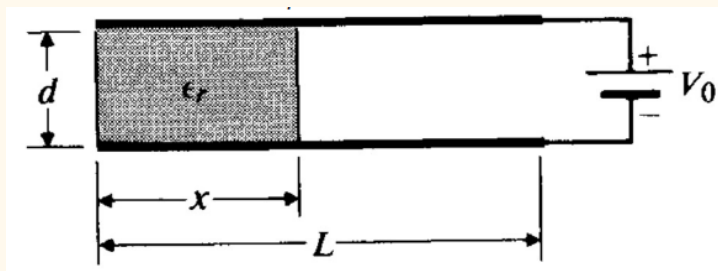
- can see easily that the current through the plates are equal

Parallel Plate Inductor in series



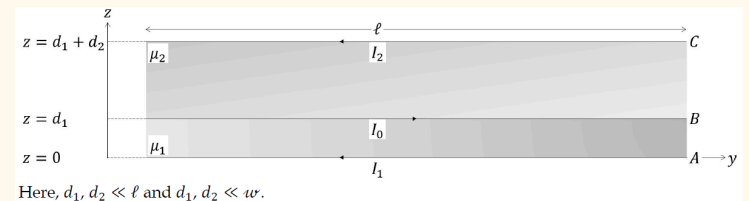
- Current through the plates are equal, although unintuitive, both are in series

Parallel Plate Capacitor in parallel



- can see easily that voltage across the two dielectrics are equal

Parallel Plate Inductor in Parallel



Here, $d_1, d_2 \ll \ell$ and $d_1, d_2 \ll w$.

- There's 3 plates, A, B, and C
- Current in B = Current in C and Current in A
- Magnetic field between BC and AC are independent (no mutual flux as plates are infinitely large)
 - Current will split in parallel inductors such that flux through both inductors are the same
- Tut2Q10 actually forces them to be in parallel as it forced the flux going through the two region to be equal, in reality, there's a distributed elements of L and C

General Steps:

General Steps:

1. Form equation of V using Laplace Equation as free charge = 0 inside the mediums, for parallel plate V varies linearly in bounded region with axis normal to plates

$$V(x) = \frac{V_2 - V_1}{d_2 - d_1} (x - d_1) + V_1$$

2. Substitute unknown with boundary voltage, V_B
3. Find E in terms of V_B
4. Boundary condition at the boundary between two electric mediums with free charge = 0 to obtain V_B
5. Substitute back to get V and E for all region
6. With E, solve boundary condition at the plates to get $\rho_{s, \text{free}}$
7. Multiply with area to get $Q_{\text{free}}, Q_{\text{free, top}} = Q_{\text{free, bot}}, Q_{\text{bound, top}} \neq Q_{\text{bound, bot}}$
8. $C = |Q|/\Delta V$
9. Double check against capacitor formulas

1. Form equation of A using Laplace Equation as free current = 0 inside the medium, for parallel plate A points in the direction of current (x) and varies linearly in bounded region with axis normal to plates (z)

$$A_x(z) = k_1 z + k_2$$

2. Find B in terms of k_1
3. Solve boundary condition at the plates with $J_{s, \text{free}} = I/w$ to get k_1
 - a. solving for $J_{s, \text{free}} = 0$ for boundary between two magnetic mediums is redundant
4. Substitute back to get B, A and k_2 can be ignored
5. $L = \Phi/I$ or $1/2 LI^2 = u_B \cdot \text{volume}$
6. Double check against inductor formulas

Preparation for Time Varying Field

Electrostatics (Time Independent)

We can describe the electrostatic field in a volume of space R by using the **electric field intensity**,

$$\mathbf{E} = \mathbf{E}(x, y, z) = -\nabla V(x, y, z)$$

Here, the **gradient** of the electric scalar potential, $V = V(x, y, z)$,

$$\nabla V(x, y, z) = \left[\frac{\partial}{\partial x} V(x, y, z) \right] \mathbf{a}_x + \left[\frac{\partial}{\partial y} V(x, y, z) \right] \mathbf{a}_y + \left[\frac{\partial}{\partial z} V(x, y, z) \right] \mathbf{a}_z$$

with $\mathbf{a}_x$, $\mathbf{a}_y$ and $\mathbf{a}_z$ unit vectors in the directions of increasing x, y and z, respectively. This is equivalent to

$$\nabla \times \mathbf{E}(x, y, z) = \mathbf{0}$$

That is, the **curl** of the electric field intensity associated with an electrostatic field is zero. This follows from ...

$$\mathbf{E} = -\nabla V \Leftrightarrow \nabla \times \mathbf{E} = \mathbf{0}$$

- Static E being the gradient of a scalar implies that the curl of static E is 0
- this shows that static E is never curling

Magnetostatics (Time Independent)

We can describe the magnetostatic field in the volume of space R by using the **magnetic flux density**,

$$\mathbf{B} = \mathbf{B}(x, y, z) = \nabla \times \mathbf{A}(x, y, z)$$

Here, the **curl** of the magnetic vector potential,

$$\nabla \times \mathbf{A}(x, y, z) = \begin{vmatrix} \mathbf{a}_x & \mathbf{a}_y & \mathbf{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x(x, y, z) & A_y(x, y, z) & A_z(x, y, z) \end{vmatrix} = \left(\frac{\partial}{\partial y} A_z - \frac{\partial}{\partial z} A_y \right) \mathbf{a}_x + \left(\frac{\partial}{\partial z} A_x - \frac{\partial}{\partial x} A_z \right) \mathbf{a}_y + \left(\frac{\partial}{\partial x} A_y - \frac{\partial}{\partial y} A_x \right) \mathbf{a}_z$$

This is equivalent to

$$\nabla \cdot \mathbf{B}(x, y, z) = 0$$

That is, the **divergence** of the magnetic flux density associated with a magnetostatic field is zero. This follows from ...

$$\mathbf{B} = \nabla \times \mathbf{A} \Leftrightarrow \nabla \cdot \mathbf{B} = 0$$

- Static B being the curl of a vector implies that the divergence of static B is 0
- this shows that static B is never diverging, there is no magnetic monopoles
 - all magnetic dipoles have the same amount of B pointing out and in, always circulating but never diverging

$$\nabla \times \mathbf{E} = -\nabla \times \nabla V = - \begin{vmatrix} \mathbf{a}_x & \mathbf{a}_y & \mathbf{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \frac{\partial V}{\partial x} & \frac{\partial V}{\partial y} & \frac{\partial V}{\partial z} \end{vmatrix} = - \left(\frac{\partial^2 V}{\partial y \partial z} - \frac{\partial^2 V}{\partial z \partial y} \right) \mathbf{a}_x - \left(\frac{\partial^2 V}{\partial z \partial x} - \frac{\partial^2 V}{\partial x \partial z} \right) \mathbf{a}_y - \left(\frac{\partial^2 V}{\partial x \partial y} - \frac{\partial^2 V}{\partial y \partial x} \right) \mathbf{a}_z = \mathbf{0}$$

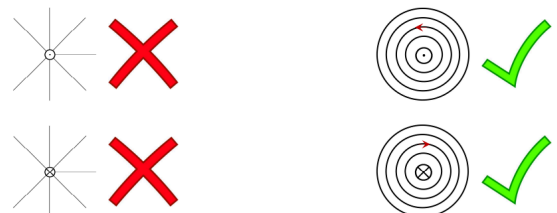
Physically,



- Static E field will not have curl
 - curving of electric field doesn't mean curl
 - curl means go around circle
 - point charge will just stop from one end to another, no going in circle
- Curl of static E is always = 0

$$\nabla \cdot \mathbf{B}(x, y, z) = \nabla \cdot [\nabla \times \mathbf{A}(x, y, z)] = \frac{\partial}{\partial x} \left(\frac{\partial}{\partial y} A_z - \frac{\partial}{\partial z} A_y \right) + \frac{\partial}{\partial y} \left(\frac{\partial}{\partial z} A_x - \frac{\partial}{\partial x} A_z \right) + \frac{\partial}{\partial z} \left(\frac{\partial}{\partial x} A_y - \frac{\partial}{\partial y} A_x \right) = 0$$

Physically,



- Static B field will not have divergence
 - always go round in circle, a close loop
- Div of static B is always = 0
 - div of curl is always = 0

- curl of gradient field is always = 0

In fact, the Poisson equation,

$$\nabla^2 V(x, y, z) = -\frac{1}{\epsilon_0} \rho(x, y, z)$$

is equivalent to the **divergence** of the electric field intensity associated with an electrostatic field,

$$\nabla \cdot \mathbf{E}(x, y, z) = \frac{1}{\epsilon_0} \rho(x, y, z)$$

Here, the **divergence** of the electric field intensity,

$$\nabla \cdot \mathbf{E}(x, y, z) = \frac{\partial}{\partial x} E_x(x, y, z) + \frac{\partial}{\partial y} E_y(x, y, z) + \frac{\partial}{\partial z} E_z(x, y, z)$$

So, with $\mathbf{E}(x, y, z) = -\nabla V(x, y, z)$, we have

$$\nabla \cdot \mathbf{E}(x, y, z) = -\frac{\partial^2}{\partial x^2} V(x, y, z) - \frac{\partial^2}{\partial y^2} V(x, y, z) - \frac{\partial^2}{\partial z^2} V(x, y, z) = -\nabla^2 V(x, y, z)$$

$$\nabla \cdot \mathbf{E} = \frac{1}{\epsilon_0} \rho$$

- when charge density of a point is +ve, the static electric field diverges away from that point
- when charge density of a point is -ve, the static electric field converges towards that point
- this shows that static E is always diverging or converging from a point

In fact, the Poisson equation,

$$\nabla^2 \mathbf{A}(x, y, z) = -\mu_0 \mathbf{J}(x, y, z)$$

is equivalent to the **curl** of the magnetic flux density associated with a magnetostatic field,

$$\nabla \times \mathbf{B}(x, y, z) = \mu_0 \mathbf{J}(x, y, z)$$

provided $\nabla \cdot \mathbf{A}(x, y, z) = 0$, which can always be achieved! Consider the x- component,

$$\begin{aligned} (\nabla \times \mathbf{B})_x &= [\nabla \times (\nabla \times \mathbf{A})]_x = \frac{\partial}{\partial y} (\nabla \times \mathbf{A})_z - \frac{\partial}{\partial z} (\nabla \times \mathbf{A})_y \\ &= \frac{\partial}{\partial y} \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right) - \frac{\partial}{\partial z} \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \right) \\ &= \frac{\partial}{\partial x} \left(\frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z} \right) - \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) A_x = [\nabla(\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A}]_x \end{aligned}$$

So, $\nabla \times \mathbf{B} = \nabla(\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A}$ and $\nabla \times \mathbf{B} = -\nabla^2 \mathbf{A}$ if $\nabla \cdot \mathbf{A} = 0$.

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} \quad (\text{iff } \nabla \cdot \mathbf{A} = 0)$$

- when current density of a point is +ve, the static magnetic field curl positively (anti-cw) around the direction of current density
- when current density of a point is -ve, the static magnetic field curl negatively (cw) around the direction of current density
- this shows that static B is always curling positively around current

The electric field intensity $\mathbf{E}$ and magnetic flux density $\mathbf{B}$ of time-independent electric and magnetic fields, respectively, satisfy the following **Maxwell equations**:

$$\nabla \cdot \mathbf{E} = \frac{1}{\epsilon_0} \rho, \quad \nabla \times \mathbf{B} = \mu_0 \mathbf{J}$$

and

$$\nabla \times \mathbf{E} = 0, \quad \nabla \cdot \mathbf{B} = 0.$$

Here, ρ is the volume charge density and $\mathbf{J}$ is the volume current density. They are time-independent too. Note that

$$\nabla \cdot \mathbf{J} = \frac{1}{\mu_0} \nabla \cdot (\nabla \times \mathbf{B}) \quad \stackrel{\text{just like}}{=} \quad 0.$$

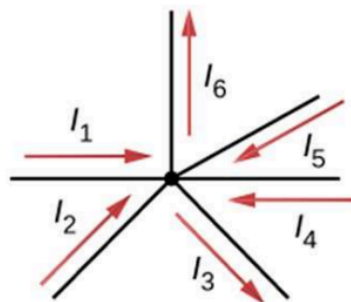
$\nabla \cdot (\nabla \times \mathbf{A}) = 0$

This is consistent with the fact that $\mathbf{J}$ is the volume density of steady current.

- Divergence of current density = 0, current is steady and non time varying

For example, Kirchhoff's junction rule:

$$\nabla \cdot \mathbf{J} = 0$$



$$\sum I_{\text{in}} = \sum I_{\text{out}} \\ I_1 + I_2 + I_4 + I_5 = I_3 + I_6$$

What happen if $\mathbf{J}$ were time-dependent, i.e., non-steady? A time-dependent $\mathbf{J}$ will produce a time-dependent $\mathbf{B}$, ...

- In steady current (conservation of charge)

- no charge accumulation in junction
- Current go in = current come out
- Current converges = current diverges
- Divergence of current = 0
- alternatively:
 - Divergence of current = divergence of curl of B = 0 (divergence of curl = 0)
- What if charge can accumulate or dissipate at a point?
 - divergence of current is non zero
 - if accumulate, charge increasing, then net flow of current density = $\text{div}(\mathbf{J}) = \nabla \cdot \mathbf{J} < 0$
 - accumulation of charge density over time, $\partial \rho / \partial t > 0$
 - if dissipate, charge decreasing, then net flow of current density = $\text{div}(\mathbf{J}) = \nabla \cdot \mathbf{J} > 0$
 - accumulation of charge density over time, $\partial \rho / \partial t < 0$
 - J is in A/m^2 , C/sm^2
 - div is derivative in distance, div J is in C/sm^3
 - $\nabla \cdot \mathbf{J} = - \partial \rho / \partial t$ (the continuity equation)
 - this equation shows conservation
 - When charge density increase, then the divergence of charge must be negative to increase the density, as charge is conserved
 - mechanical equivalent would be when mass density increase, then the divergence of mass must be negative to increase the density, as mass is conserved